

AIG130 – Lab 5

Objective

This lab is to explore Automated Machine Learning (AutoML) platforms. You will explore how AutoML simplifies the process of applying machine learning models to solve problems.

Group Formation

Students can work individually or form groups of up to 4 members. Collaboration and effective communication are encouraged for those working in groups.

Assignment

- Part 1: Choose an AutoML platform (e.g., Google AutoML, AWS AutoGluon, or Microsoft Azure AutoML) and explore its features and capabilities.
- Part 2: You have two options: 1) Choose one of your works from AIG100 and re-implement it with AutoML. 2) Choose a new dataset and implement your choice of machine learning model with AutoML.

Deliverables

Report: Submit a PDF report covering all the steps you took to complete both tasks. Consider this report as a tutorial for completing this lab!

Video Demo: Create a video showcasing your completed work with all parts of your work clearly shown in the video.

Individual Reflection

Each student must submit a separate reflection document answering the following:

1. If you worked individually, why did you choose to go solo? What challenges did you encounter?
2. If you did it as a group, how were tasks distributed, and what was your specific task? What challenges did you encounter?
3. What lessons did you learn while working on this lab? What would you do differently if you were asked to do this lab again?

Submission

Submissions are done through the provided links in Blackboard. There are three links:

1. Group report (one submission per group): 4%
2. Individual reflection: 2%
3. Demo file/link: 2%

Assessment Criteria

- Quality and depth of research and analysis.

- Clarity and organization of the report.
- Quality and completeness of the demo video.
- Thoughtfulness of the individual reflection.